

Childcare Affordability & Workforce Participation Across U.S. Counties

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DSC 640 Data Presentation and Visualization

May 7, 2026

Summary of Analysis

Findings

This project analyzed childcare affordability as both a household budget issue and a workforce participation issue for families with young children. Across Milestone 1 and Milestone 3, the strongest pattern was that infant care is the highest-cost category. That affordability burden depends on the relationship between price and household income, not price alone. The 7% benchmark was used as a practical decision threshold (Child Care Aware of America, n.d.-b). This framing kept the analysis focused on decision usefulness rather than on descriptive statistics alone.

Across the final deliverables, four findings guided the story. First, childcare prices increased substantially, with trend visuals showing costs rising from about \$145 per week in 2012 to \$238 in 2023 (roughly 60%). Second, age-group differences matter: infant care was consistently higher than toddler and preschool care. Third, the affordability burden varied sharply by income context, so areas with similar prices could have very different burden ratios once income was included. Fourth, higher burden categories aligned with lower maternal workforce participation, including an approximately 11-point gap between low- and high-burden groups in the project narrative.

Assumptions

Weekly childcare costs were annualized over 52 weeks for affordability calculations. Median values were used as representative values for county-level or grouped summaries. Benchmark comparisons relied on the 7% affordability threshold. In early exploration, imputed observations were included when flagged and interpreted as directional rather than definitive. These assumptions made cross-region and cross-year comparisons possible within a single framework.

Items Still Needing Clarification

Three questions remain for future refinement: whether imputed years should be moved to sensitivity-only views, whether county rollups should be explicitly weighted by population or by children under 5, and whether separate recommendations are needed for full-time versus before- and after-school care. Clarifying these areas would strengthen methodological transparency and improve reproducibility.

Direction of Story

The core message is that affordability, not raw price, should guide family choices and policy design. The story moves through four stages: frame rising costs and age-group burden; introduce the affordability ratio logic; connect burden to workforce outcomes; and end with role-specific actions for parents, employers, and policymakers. This sequence keeps the narrative practical by moving from evidence to decisions, and it helps audiences see that individual choices and structural constraints operate simultaneously.

Target Audience

The primary audience is new mothers and caregivers considering childcare and returning to work during the early parenthood period. Secondary audiences include employers, community organizations, and policy advocates who influence benefit access, subsidy awareness, and local childcare support systems. This audience strategy reflects the reality that caregiving decisions are made at the household level, but barriers are often systemic.

Mediums

Three mediums were used: a presentation, an infographic, and a blog post. The presentation supports guided live explanation and Q&A, the infographic supports quick scanning and sharing for busy caregivers, and the blog supports deeper asynchronous learning for readers who want fuller context. Together, these formats improve reach and retention by matching message depth to audience needs and offering both short-form and long-form pathways to the same key takeaways.

Design Decisions

Design choices prioritized clarity and low cognitive load. The visual hierarchy emphasized the most decision-relevant indicators: weekly age-group costs, the affordability ratio relative to the 7% benchmark, and workforce participation differences. A consistent navy-and-orange palette and plain-language annotations were used across

outputs to improve readability. Medium-specific visual uniqueness was also intentional, so each deliverable used distinct charts tailored to its communication purpose rather than duplicating the same visuals everywhere.

Ethical Considerations

Changes Made to the Data

The data were cleaned for missing values, standardized to comparable units, and transformed into affordability measures such as annual childcare costs and childcare costs as a percentage of income. These changes were necessary to support consistent cross-county and cross-year comparisons, but they should be described clearly so readers understand how the final measures were produced.

Legal or Regulatory Guidelines

This project relied on public, aggregate data rather than personally identifiable information, which reduces privacy and compliance concerns. Even so, the paper still needs to follow basic ethical data-use practices such as accurate citation, faithful representation of source findings, and avoiding claims that go beyond the evidence.

Risks Created by Transformations or Presentation

One ethical risk is that summarized visuals may make patterns appear more certain or more universal than they really are. Another risk is that readers may interpret the relationship between affordability and workforce participation as causal when the analysis supports only association.

Assumptions and Filtering

The analysis assumes that annualizing weekly childcare costs over 52 weeks and using the 7% benchmark provides a reasonable affordability framework. Any excluded, imputed, or summarized data should be identified clearly so readers can see what was included, what was limited, and where interpretation should remain cautious.

Data Sourcing and Verification

Credibility was supported by using established sources and by checking core claims across multiple references rather than relying on a single source. Benchmark guidance

from Child Care Aware of America and supporting context from related childcare sources helped keep the narrative grounded in publicly available evidence (Child Care Aware of America, n.d.-a; Child Care Aware of America, n.d.-b; TOOTRiS, n.d.).

Ethical Data Acquisition

The data were acquired ethically because they came from public, non-sensitive sources intended for public use and analysis. No personal records, confidential household information, or restricted datasets were used in this project.

Mitigating Ethical Implications

These ethical concerns can be reduced by documenting transformations, labeling assumptions, and clearly distinguishing correlation from causation throughout the paper and visuals. It is also important to note source limitations, so the findings are presented as decision-support evidence rather than absolute conclusions.

Lessons Learned

What I Would Do Differently Next Time

If I were to do this project differently next time, I would add weighting and uncertainty treatment earlier in the pipeline and formalize a sensitivity appendix so methodological choices are visible without overloading the main narrative. I would also run additional audience testing before finalizing the visual hierarchy to improve clarity of communication.

What I Enjoyed the Most

What I enjoyed most about this project was translating analysis into practical, audience-specific actions. Building a connected story across presentation, infographic, and blog made the project feel useful beyond class requirements. The process reinforced the role of data storytelling in fostering empathy and supporting evidence-based decision-making.

References:

Child Care Aware of America. (n.d.-a). Price landscape 2024.

<https://www.childcareaware.org>

Child Care Aware of America. (n.d.-b). Cost of child care.

<https://www.childcareaware.org/families/cost-child-care/>

TOOTRiS. (n.d.). Child care in Texas can cost up to \$10,000.

<https://tootris.com/edu/blog/parents/child-care-in-texas-can-cost-up-to-10000/>

OpenAI. (2026). ChatGPT (Apr 25 version) [Large language model].

<https://chat.openai.com/>